

We can create a list for the 'Mazda RX4' values, and define a class 'Car' for the same as shown below:

```
> m <- list(mpg=21.0, cyl=6, disp=160)
> class(m) <- "Car"
> m
$mpg
[1] 21
$cyl
[1] 6
$disp
[1] 160
attr(,"class")
[1] "Car"
```

We can also add the 'Hornet 4 Drive' attributes to a new instance, as follows:

```
> h <- list(mpg=21.4, cyl=6, disp=258)
> class(h) <- "Car"
> h
$mpg
[1] 21.4

$cyl
[1] 6

$disp
[1] 258

attr(,"class")
[1] "Car"
```

The *attributes()* function returns the object's attributes, as shown below:

```
> attributes(h)
$names
[1] "mpg" "cyl" "disp"

$class
[1] "Car"
```

You can also add attributes to an object using the *attr()* function. For example:

```
> attr(m, "hp") <- c(110)

> print(m)

$mpg
[1] 21
```

```
$cyl
[1] 6

$disp
[1] 160

attr(,"hp")
[1] 110
```

We can define a method to operate on the class object, as follows:

```
> print.Car <- function(obj) {
+   cat("mpg: ", obj$mpg, "\n")
+   cat("cyl: ", obj$cyl, "\n")
+   cat("disp: ", obj$disp, "\n")
+ }

> print(m)
mpg: 21
cyl: 6
disp: 160
```

A new Gear class can be created with simple inheritance from the Car class, as shown below:

```
> g <- list(mpg=21, cyl=6, disp=160, gear=4)

> class(g) <- c("Gear", "Car")

> inherits(g, "Car")
[1] TRUE

> print(g)
mpg: 21
cyl: 6
disp: 160
```

Another example for S3 classes is the time series 'ts' objects. Consider the consumer prices (annual %) inflation data for India from July 2021 to June 2022. The *ts()* function can be used to create the time-series objects as illustrated below:

```
> cpi <- ts(data=c(162.5, 162.9, 163.2, 165.5, 166.7, 166.2,
165.7, 166.1, 167.7, 170.1, 171.7, 172.6))

> cpi
Time Series:
Start = 1
End = 12
Frequency = 1
[1] 162.5 162.9 163.2 165.5 166.7 166.2 165.7 166.1 167.7
```